

HCI History and Today's Opportunities – What We Anticipated, What We Did Not

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ABSTRACT

Human-computer interaction has entered a third, globally-connected era. Visions that drove research and development in the first era were realized in the second, with CHI a key player. The third presents opportunities and a need for creativity to address challenges. HCI has drawn on computer science, human factors, information systems, and information science. It relies on design and interacts with AI. Students, practitioners, and academics can gain an understanding of forces that have guided the interaction of related disciplines, constraints imposed by human nature, trajectories we are following, and opportunities and issues that will engage us in the years ahead.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction**; • **Social and professional topics**; • **professional topics**; • **History of computing**; • **History of software**, **History of people**, **History of hardware**;

KEYWORDS

HCI, Human Factors, Information Systems, Information Science, Design, AI, History, Future

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1 BENEFITS OF UNDERSTANDING HCI HISTORY AND ITS RELATIONSHIP TO CHALLENGES TODAY

Participants will learn how and why the disciplines that have contributed to HCI over time differ in their objectives and methods. They will get a deeper sense of how human nature can constrain technology uses, some important consequences that were not anticipated, and opportunities and challenges that our trajectories carry us toward.

It was said that those who do not learn history are doomed to repeat it. This is not so often the case in our rapidly changing

technological world. Fifteen years ago a new HCI rose from the ashes of the Internet bubble. Nevertheless, we still too often reinvent things that were tried and failed. Recurring patterns of success as well as failure often result from understanding or neglecting constraints imposed by human cognitive, social and organizational behavior.

The past few years have seen major changes in our interactions with devices and software, and growing uneasiness about some effects of uses and the organizations that control them. As billions of people rely ever more on technology, unintended consequences can have profound effects that can be difficult to anticipate or steer. By looking back to find patterns in what was not anticipated before, we can better anticipate and design for the future.

2 INTENDED AUDIENCE

A researcher from any field who has a focus on technology design and use can benefit by identifying branches of HCI most likely to provide appropriate insights. Students who would like to position their work, learn from the past, and identify considerations that have emerged and deserve attention can benefit. I started as a practitioner and was motivated to explore HCI history to better understand things I learned the hard way that can still be useful for others, either directly or by analogy. Everyone can profit from understanding tradeoffs that have shaped the HCI trajectory onto which they have jumped.

3 PREREQUISITES AND HANDS-ON ACTIVITY

There are no prerequisites or hands-on work, though if technology supports it there will be an opportunity for questions and discussion.

4 COURSE STYLE AND CONTENT

This 80-minute virtual lecture course will invite questions or comments prior to and during the course, in a manner to be determined. My slides rely heavily on timelines, images, and quotations, minimizing bullet points. Previous presentations followed the outline of my 2017 book, *From Tool to Partner: The Evolution of Human-Computer Interaction*. This course will draw from that material and include recent observations that cast a new light on key human-computer interaction issues that have only recently become of critical importance.

The course starts with the ongoing challenge of HCI arising from rapid changes in technology that can collide with human nature, a familiar observation that deserves a careful look. The course surveys HCI history across fields that had significant engagement. The history field of human-computer interaction falls into three eras: Prior to 1980, when computer processing was very expensive

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but some envisioned the future potential, from 1980 to 2005 when lower cost led to the rapid expansion of discretionary use and the realization of the early visions, and the fundamentally new era of HCI that presents opportunities and challenges not previously imaginable. Both transitions were disruptive, with a central role for HCI.

Prominent issues with a long history include the relationship of AI and HCI, and within HCI the balance of quantitative and qualitative approaches. More recent arrivals are the relationship of interaction design and visual design, and the diversity of customer and user goals. A longstanding challenge has been the role of digital technologies in supporting and disrupting communities. A topic moving front and center with surprising ferocity is the proliferation of unintended negative consequences of our globe-spanning work, in the hands of both the well-intentioned and the increasingly powerful bad actors, with many of us operating in a gray spectrum in between.

The historical review covers the fields of human factors, management information systems, computer science, and information science. It draws from articles and chapters in *Annals of the History of Computing*, *AI Magazine*, *CACM*, *IEEE Computer*, several

handbooks, and an *Interactions* column that I wrote and edited for several years.

5 INSTRUCTOR BACKGROUND

I have attended and been involved with CHI since it began, contributing to methods and especially to the use of collaboration technologies in organizations. Prior to joining Microsoft Research, I was Professor of Information and Computer Science at UC Irvine. I am an ACM Fellow and CHI Academy member. My recent work has focused on education, AI technology, and a collaboration with a team studying novel crowdsourcing approaches in China.

6 RESOURCES

My web site is jonathangrudin.com. My book *From Tool to Partner: The Evolution of Human-Computer Interaction* is available through Morgan Claypool site license at many universities. For several years I edited a series of essays on HCI history for *Interactions* magazine. Although my only recent blog post for *Interactions* was on lessons learned in my early research with streaming media prototypes for higher education that could be applied to remote learning forced by COVID, my blog posts for the magazine captured a wide range of observations that could be useful.